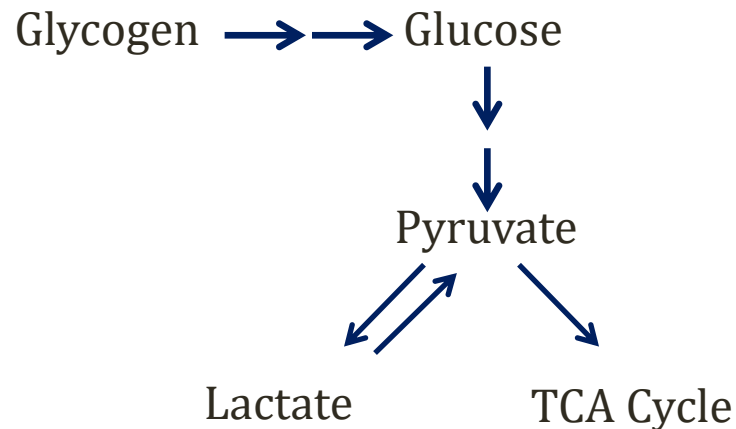


Exercise and Starvation

Jason Ryan, MD, MPH

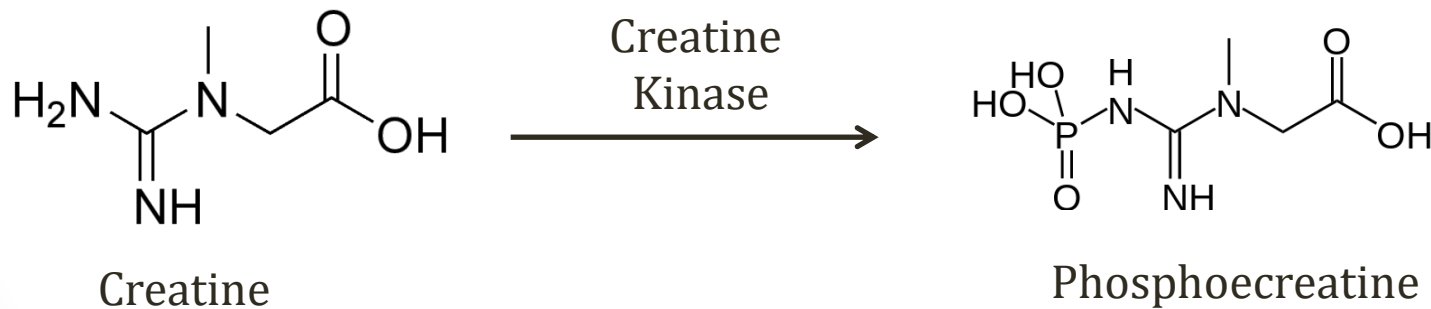
Exercise

- Rapidly depletes ATP in muscles
- Duration, intensity depends on other fuels
- Glycogen → Glucose → TCA cycle available **but slow**
- Short term needs met by creatine



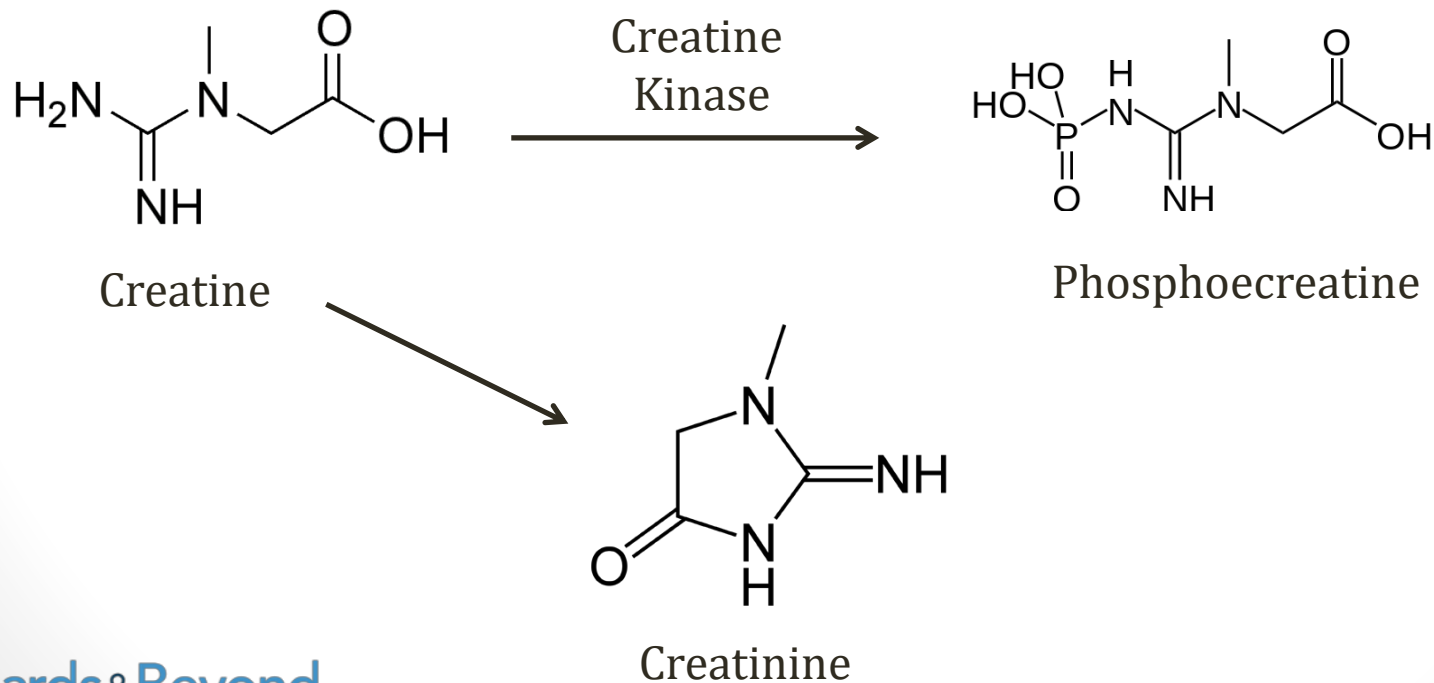
Creatine

- Present in muscles as **phosphocreatine**
- Source of phosphate groups
- Important for heart and muscles
- Can donate to ADP → ATP
- Reserve when ATP falls rapidly in early exercise



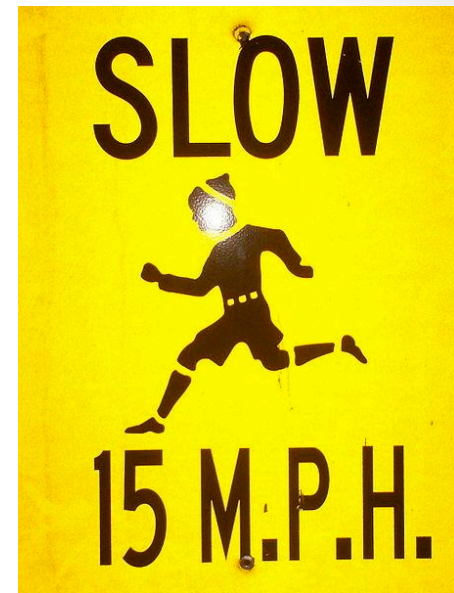
Creatinine

- Spontaneous conversion creatinine
- Amount of creatinine proportional to muscle mass
- Excreted by kidneys



ATP and Creatine

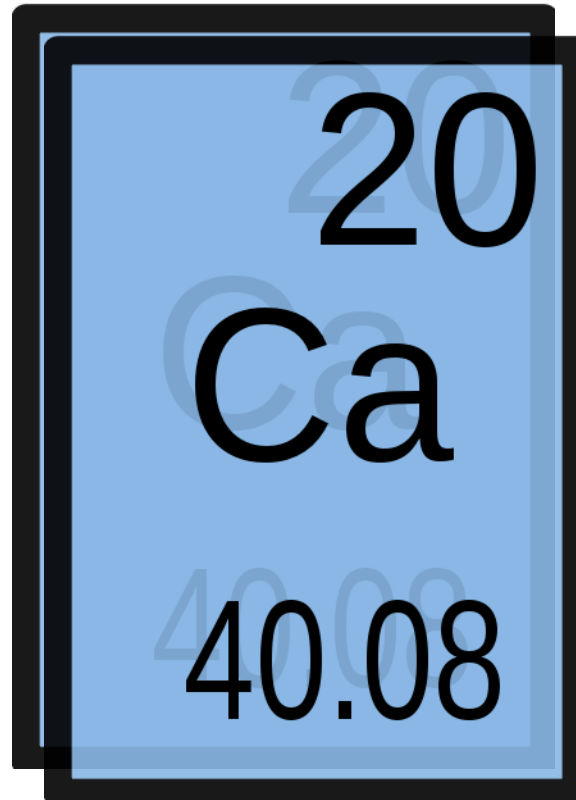
- Consumed within seconds of exercise
- Used for short, intense exertion
 - Heavy lifting
 - Sprinting
- Exercise for longer time requires other pathways
- Slower metabolism
- Result: Exercise intensity diminishes with time



Wikipedia/Public Domain

Calcium and Exercise

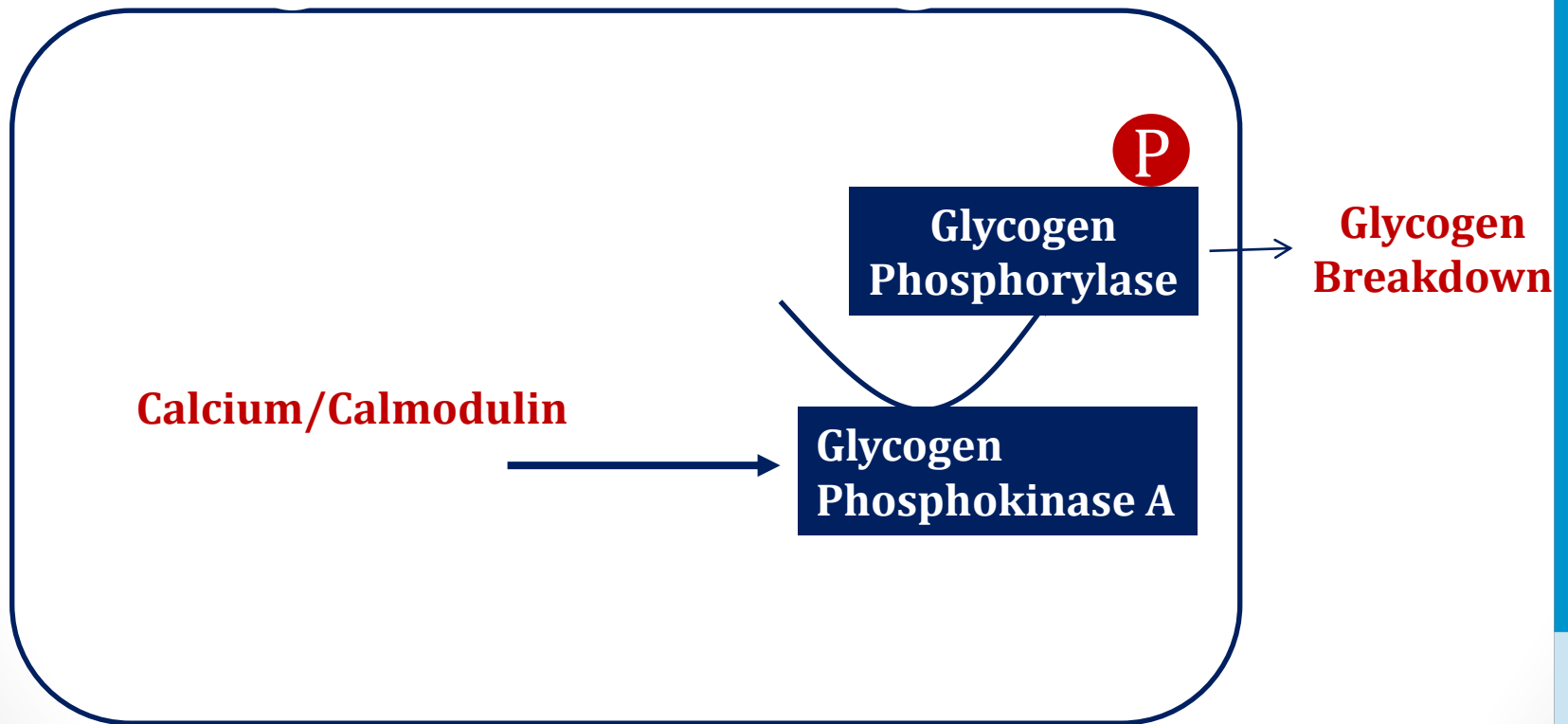
- Calcium release from muscles stimulates metabolism
- Activates glycogenolysis
- Activates TCA cycle



Me/Wikipedia

Calcium Activation

Glycogen Breakdown

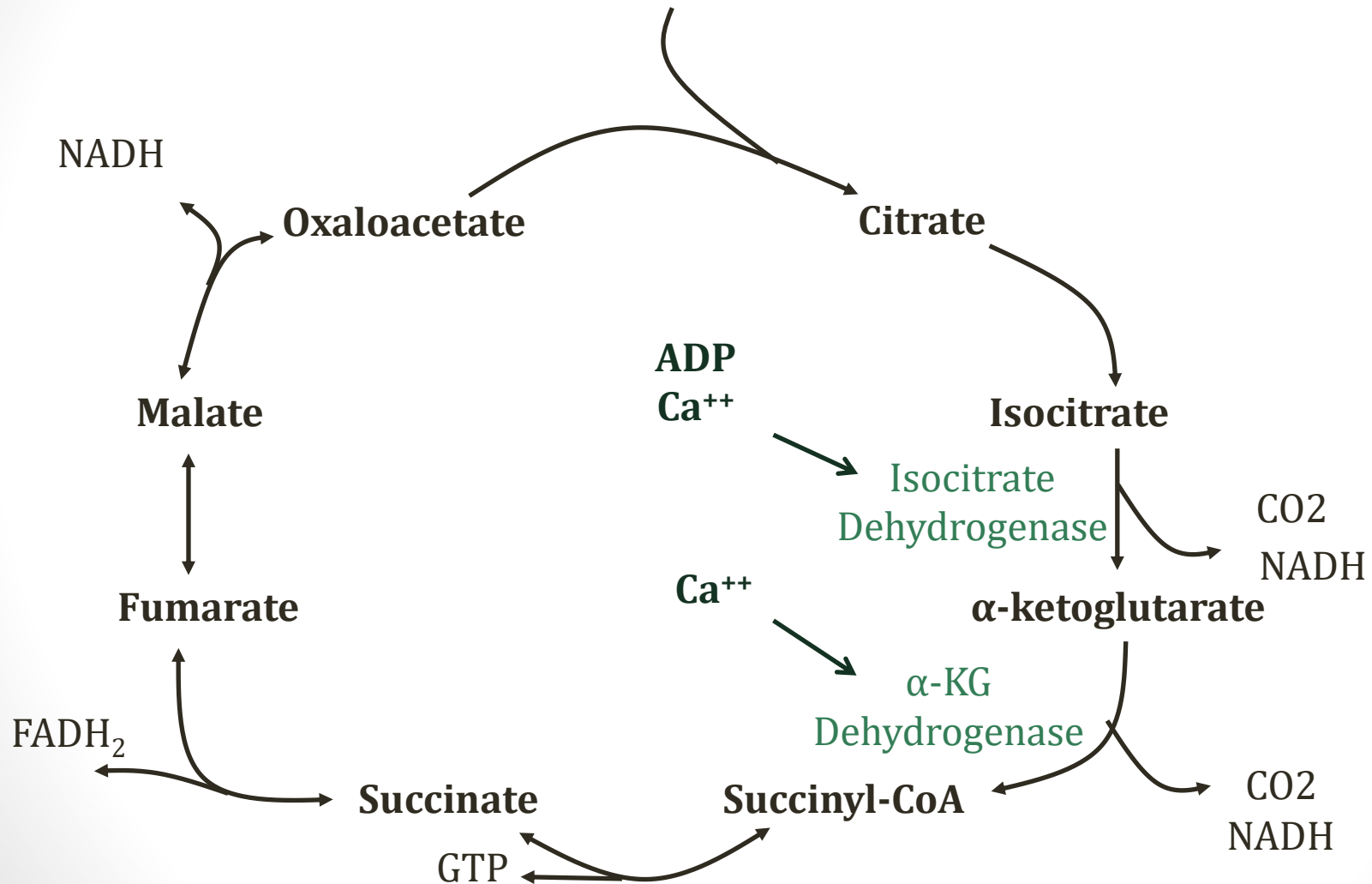


Calcium Activation

TCA Cycle

Acetyl-CoA

Pyruvate

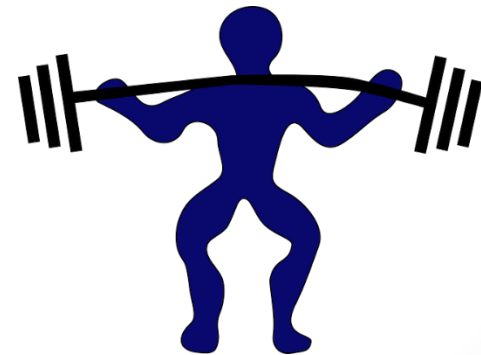


Types of Exercise

- Aerobic exercise
 - Long distance running
 - Co-ordinated effort by organ systems
 - Multiple potential sources of energy
- Anaerobic exercise
 - Sprinting, weight lifting
 - Purely a muscular effort
 - Blood vessels in muscles compressed during peak contraction
 - Muscle cells isolated from body
 - Muscle relies on it's own fuel stores



"Mike" Michael L. Baird/Wikipedia



Pixabay/Public Domain

Anaerobic Exercise

40-yard sprint

- ATP and creatine phosphate (consumed in seconds)
- Glycogen
 - Metabolized to lactate (anaerobic metabolism)
 - TCA cycle too slow
- Fast pace cannot be maintained
 - Creatinine phosphate consumed
 - Lactate accumulates

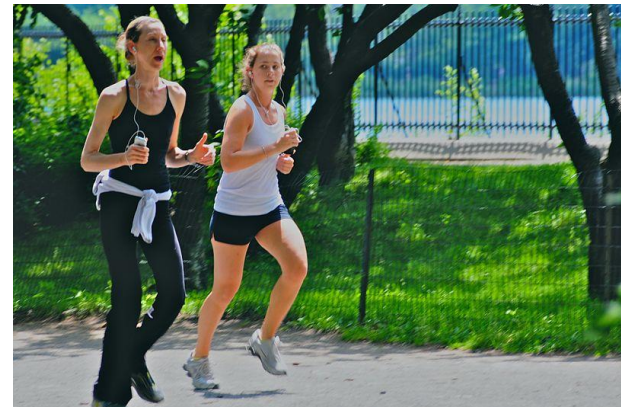


William Warby/Flickr

Moderate Aerobic Exercise

1-mile run

- ATP and creatine phosphate (consumed in seconds)
- Glycogen: metabolized to CO_2 (aerobic metabolism)
- Slower pace than sprint
 - Decrease lactate production
 - Allow time for TCA cycle and oxidative phosphorylation
- “Carbohydrate loading” by runners
 - Increases muscle glycogen content



Ed Yourdon/Wikipedia

Intense Aerobic Exercise

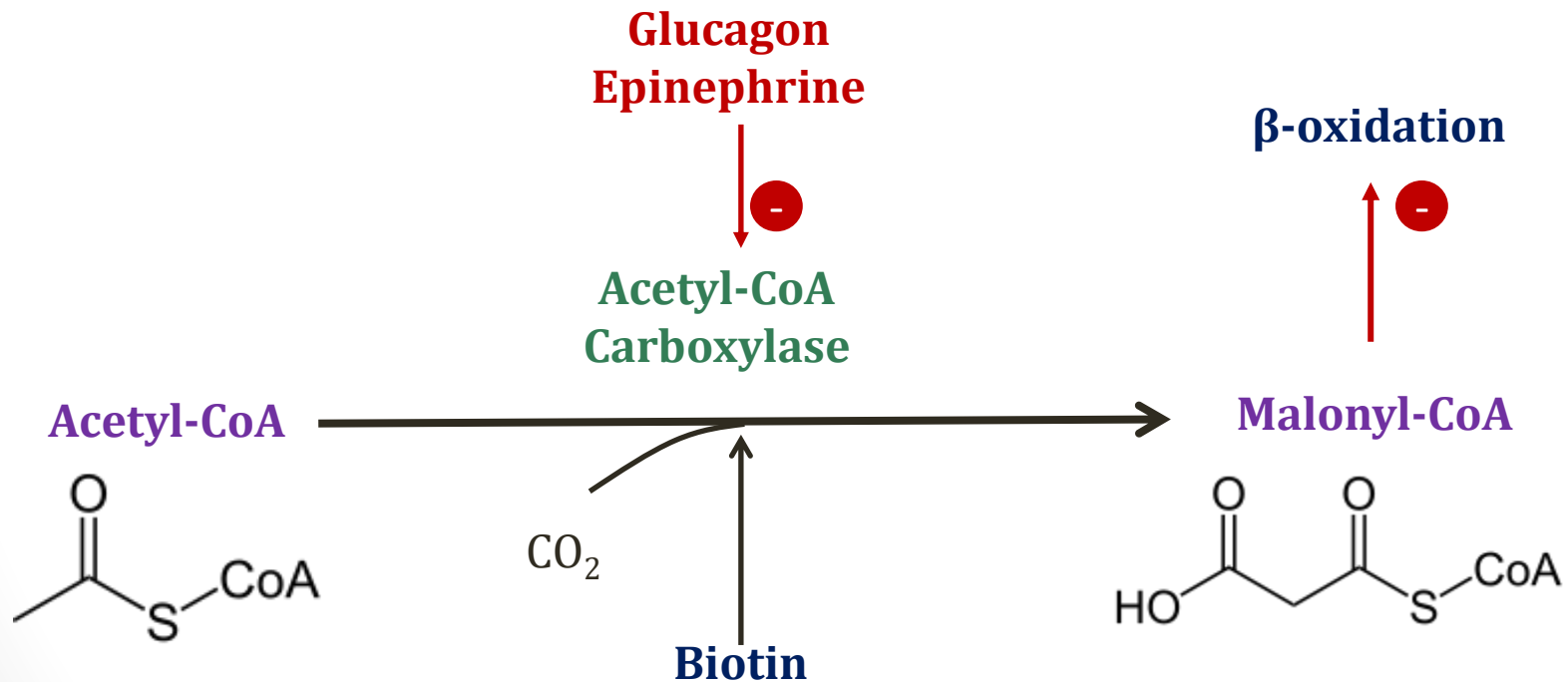
Marathon

- Co-operation between muscle, liver, adipose tissue
- ATP and creatine phosphate (consumed in seconds)
- Muscle glycogen: metabolized to CO_2
- Liver glycogen: Assists muscles → produces glucose
- Often all glycogen consumed during race
- Conversion to metabolism of fatty acids
 - Slower process
 - Maximum speed of running reduced
- Elite runners condition to use glycogen/fatty acids

Intense Aerobic Exercise

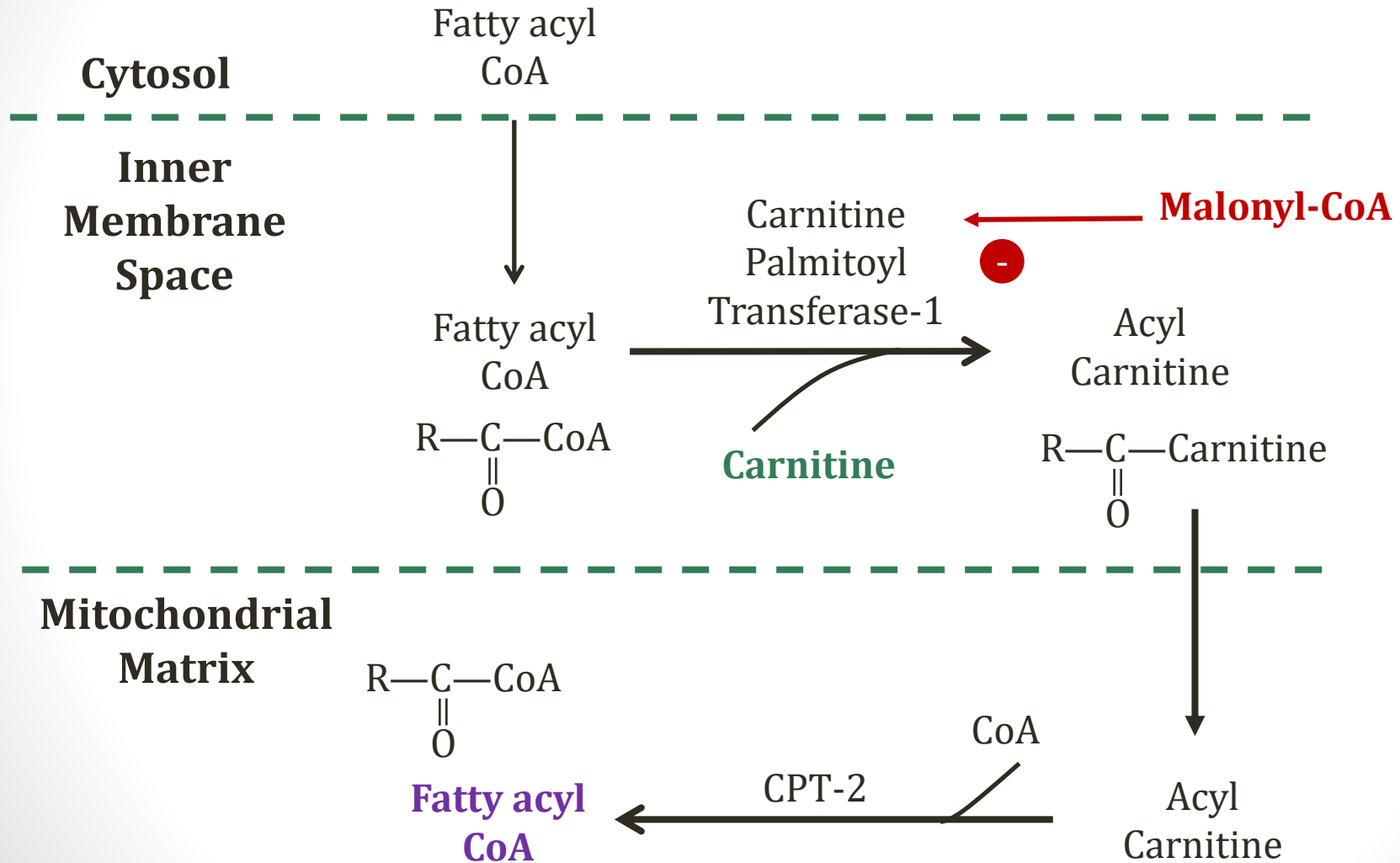
Fatty Acid Metabolism

- Malonyl-CoA levels fall



Intense Aerobic Exercise

Fatty Acid Metabolism



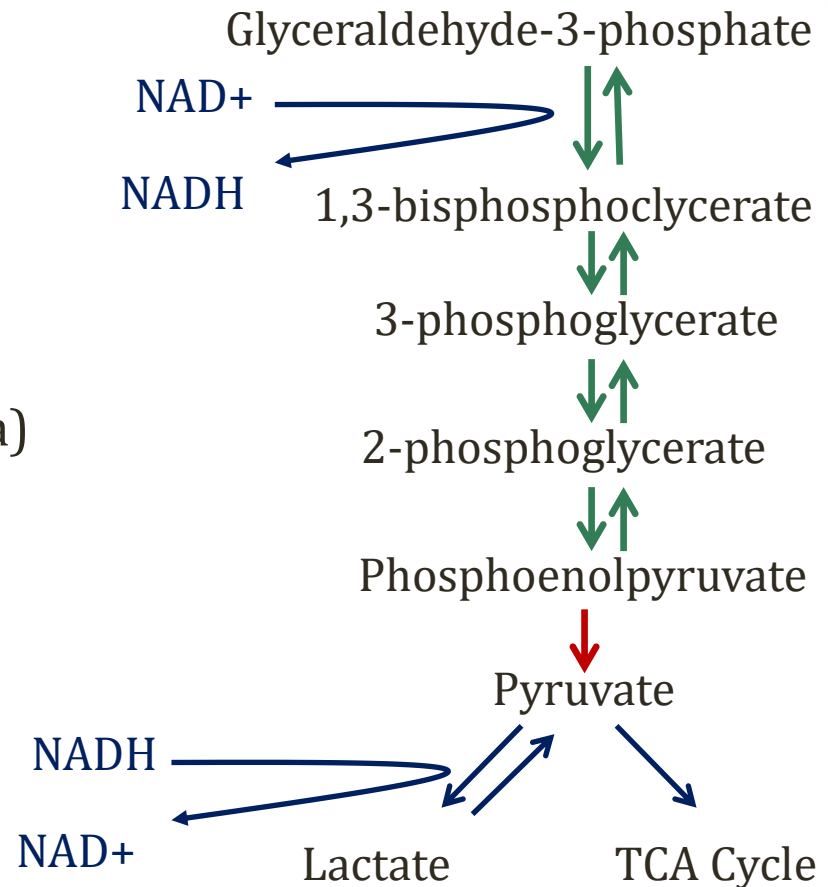
Muscle Cramps

- Too much exercise → ↑ NAD consumption
 - Exceed capacity of TCA cycle/electron transport
 - Elevated NADH/NAD ratio
- Favors pyruvate → **lactate**
- pH falls in muscles → cramps
- Distance runners: lots of mitochondria
 - Bigger, too



Muscle Cramps

- Limited supply NAD^+
- Must regenerate
- O_2 present
 - $\text{NADH} \rightarrow \text{NAD}$ (mitochondria)
- O_2 absent
 - $\text{NADH} \rightarrow \text{NAD}^+$ via LDH



Fed State

- Glucose, amino acids absorbed into blood
- Lipids into chylomicrons → lymph → blood
- **Insulin secretion**
 - Beta cells of pancreas
 - Stimulated by glucose, parasympathetic system



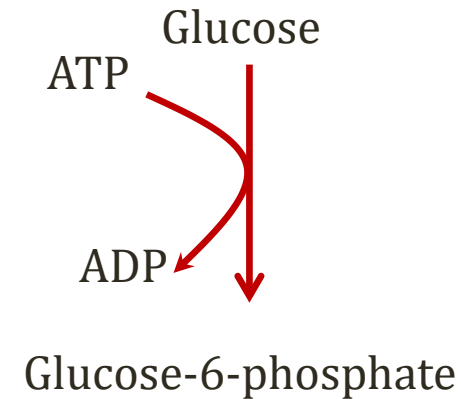
Pixabay/Public Domain

Insulin Effects

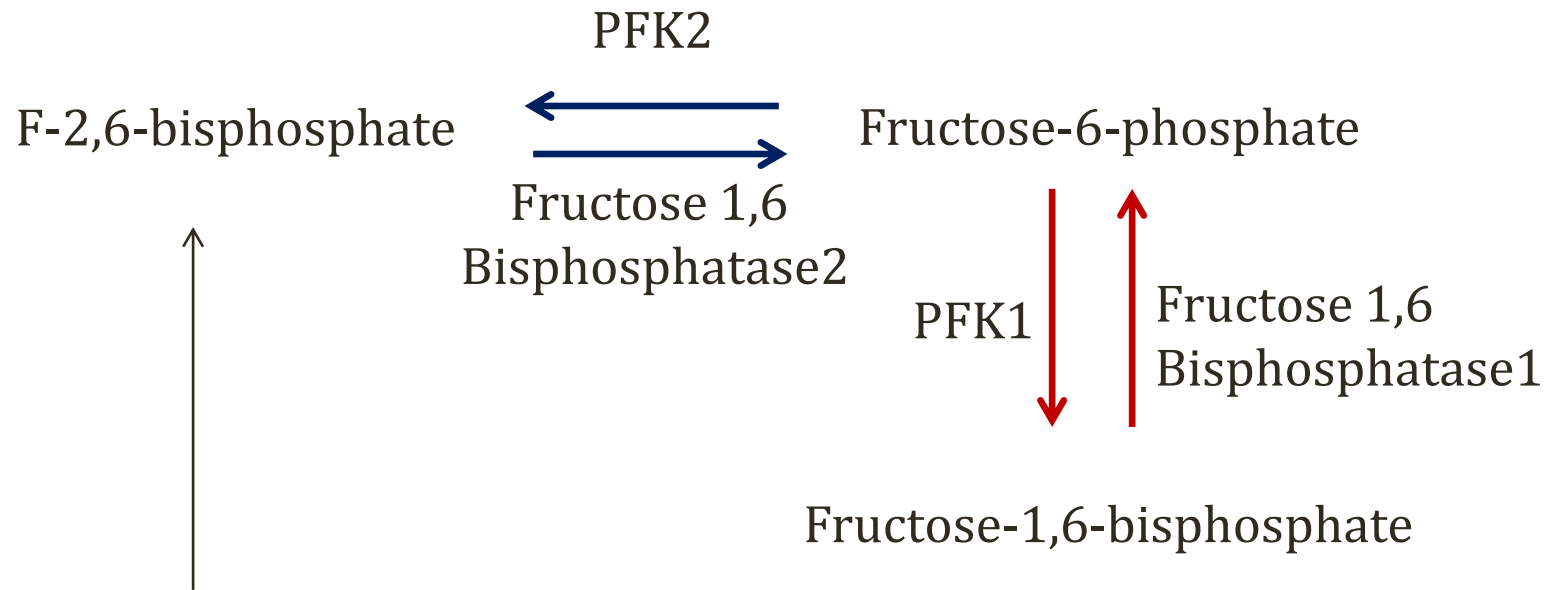
- Glycogen synthesis
 - Liver, muscle
- Increases glycolysis
- Inhibits gluconeogenesis
- Promotes glucose → adipose tissue
 - Used to form triglycerides
- Promotes uptake of amino acids by muscle
- Stimulates protein synthesis/inhibits breakdown

Insulin in the Liver

- Glucokinase
 - Found in liver and pancreas
 - **Induced by insulin**
 - Insulin promotes transcription



Insulin and Glycolysis

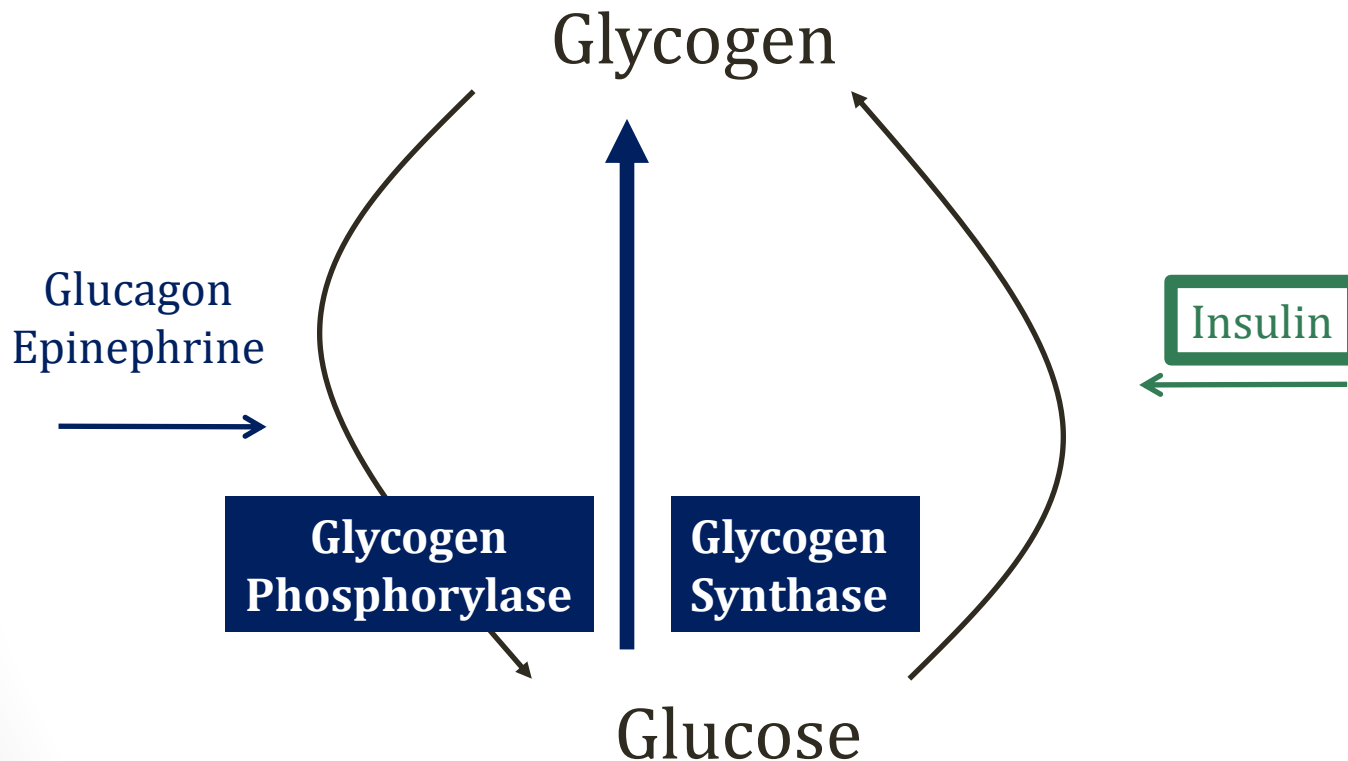


On/off switch glycolysis

↑ = glycolysis (on)

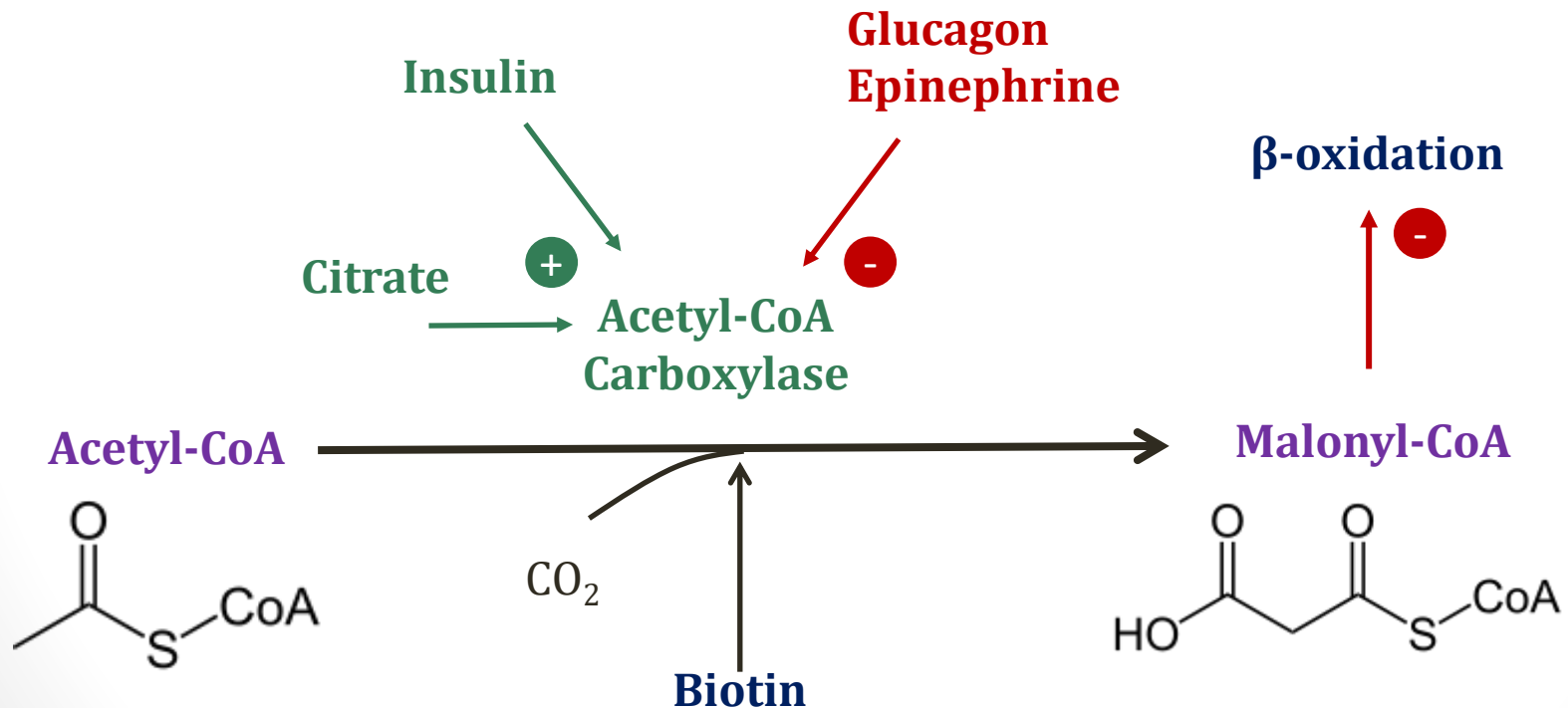
↓ = no glycolysis (gluconeogenesis)

Insulin and Glycogen



Insulin and Fatty Acids

- Acetyl-CoA converted to malonyl-CoA
- Rate limiting step



Fasting/Starvation

- Glucose levels fall few hours after a meal
- Decreased insulin
- Increased glucagon

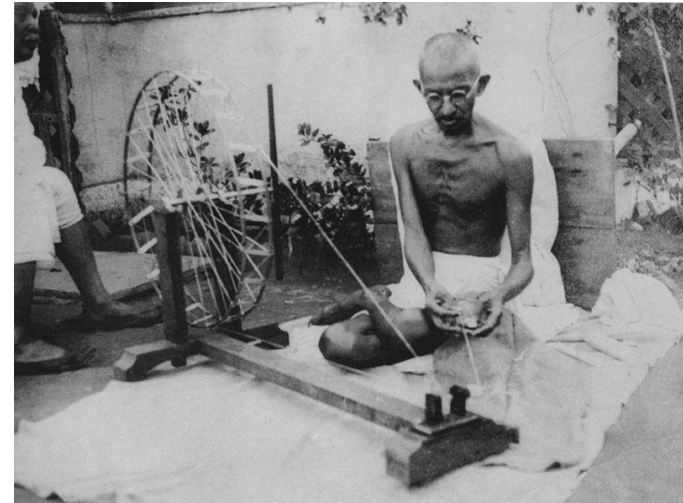
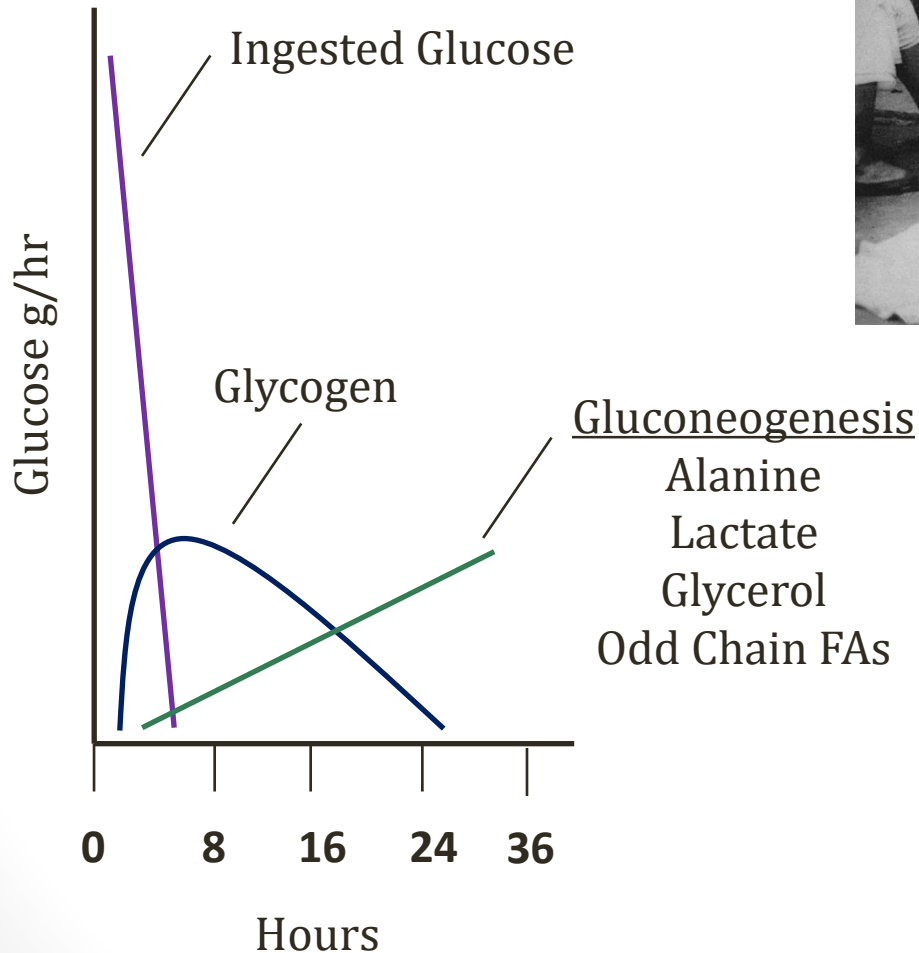


Aude/Wikipedia

Fasting/Starvation

- Key effect of glucagon
 - **Glycogen breakdown** in liver
 - Maintains glucose levels in plasma
 - **Dominant source** glucose between meals
- Other effects
 - Inhibits fatty acid synthesis
 - Stimulates release of fatty acids from adipose tissue
 - Stimulates gluconeogenesis

Glucose Sources



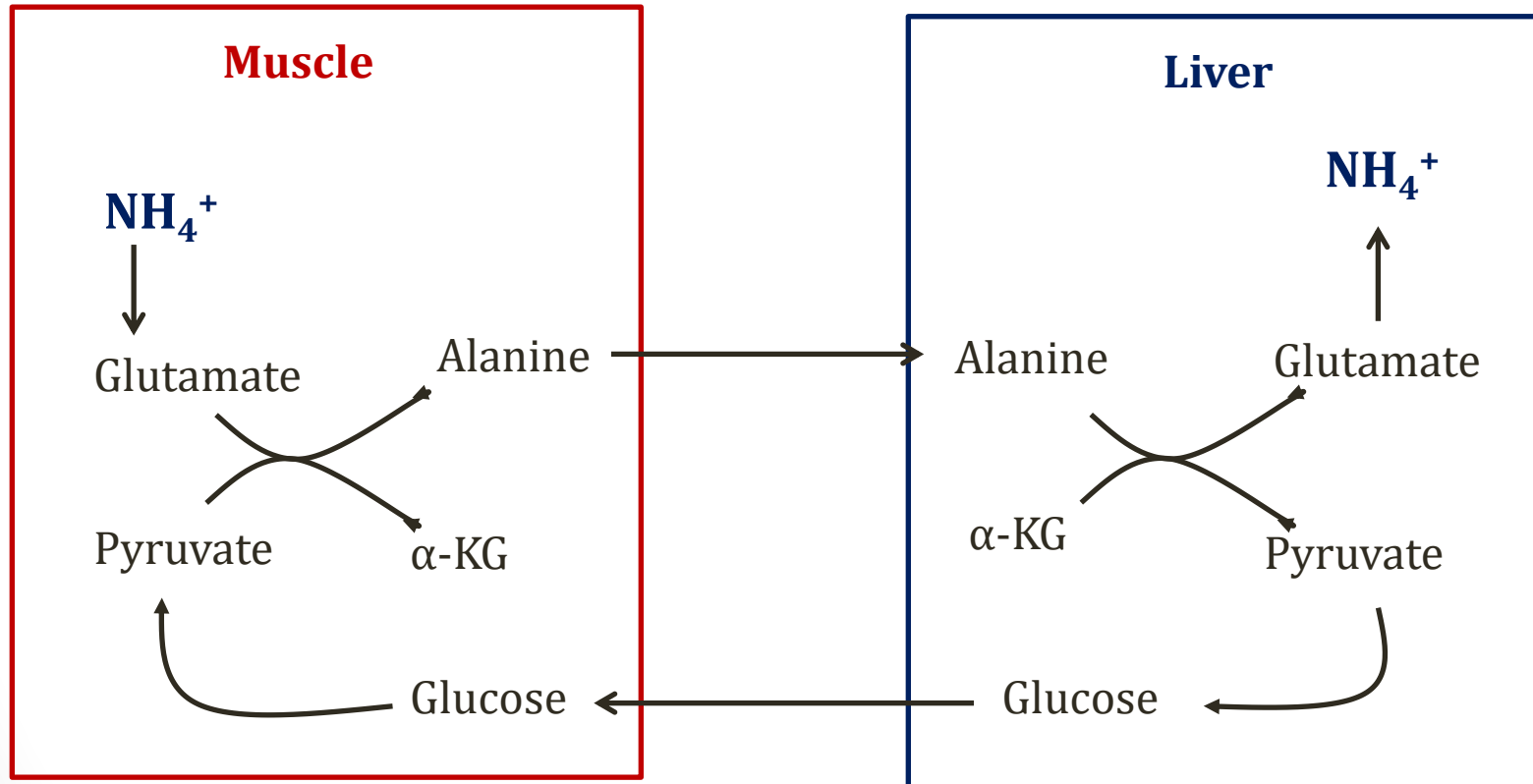
Wikipedia/Public Domain

Key Point #1:
Glycogen exhausted
after ~24 hours

Key Point #2:
Glucose levels maintained
in fasting by **many sources**

Starvation

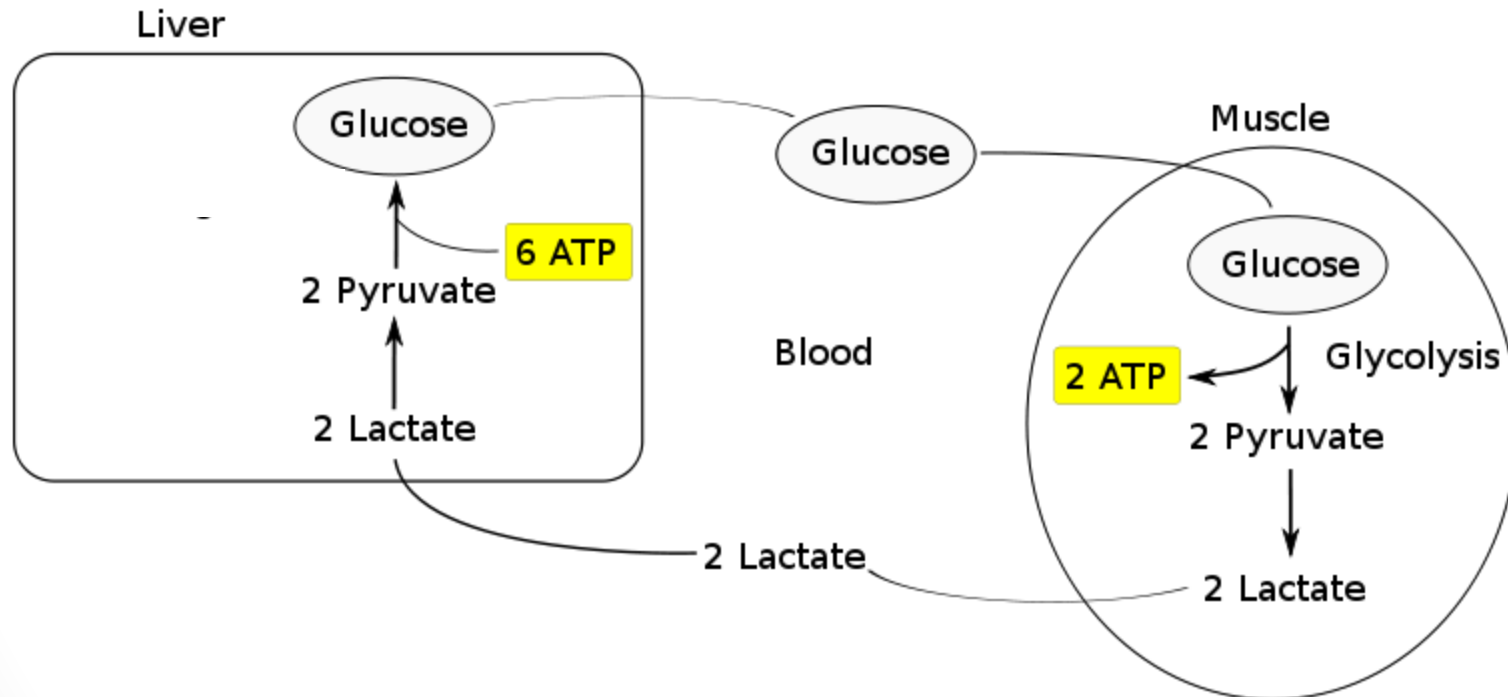
Alanine Cycle



Key Point:
Peripheral tissue alanine $\rightarrow$ glucose

Starvation

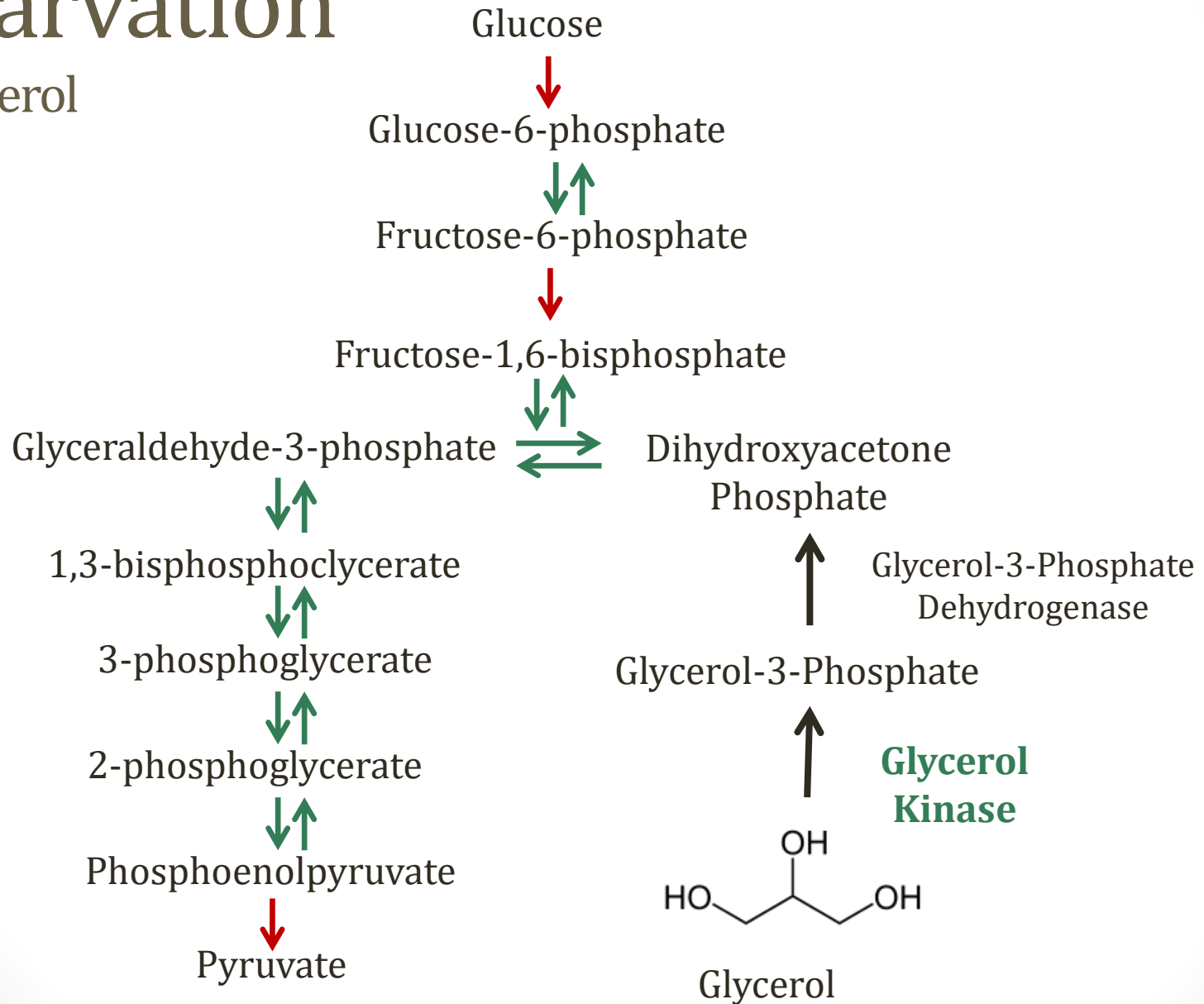
Cori Cycle



Petaholmes/Wikipedia

Starvation

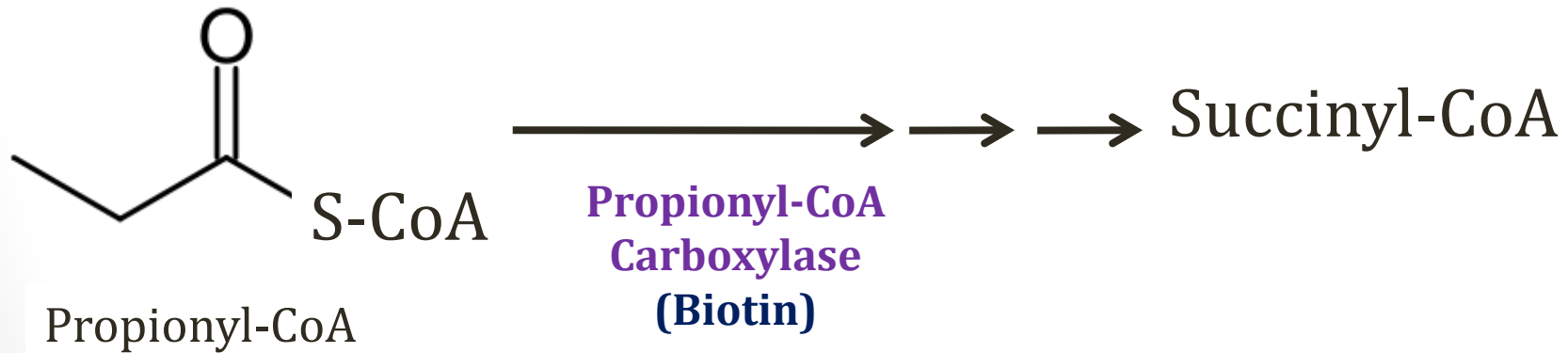
Glycerol



Starvation

Odd Chain Fatty Acids

- β -oxidation proceeds until 3 carbons remain
- Propionyl-CoA $\rightarrow$ Succinyl-CoA $\rightarrow$ TCA cycle
- Key point: Only odd chain FA $\rightarrow$ **glucose**



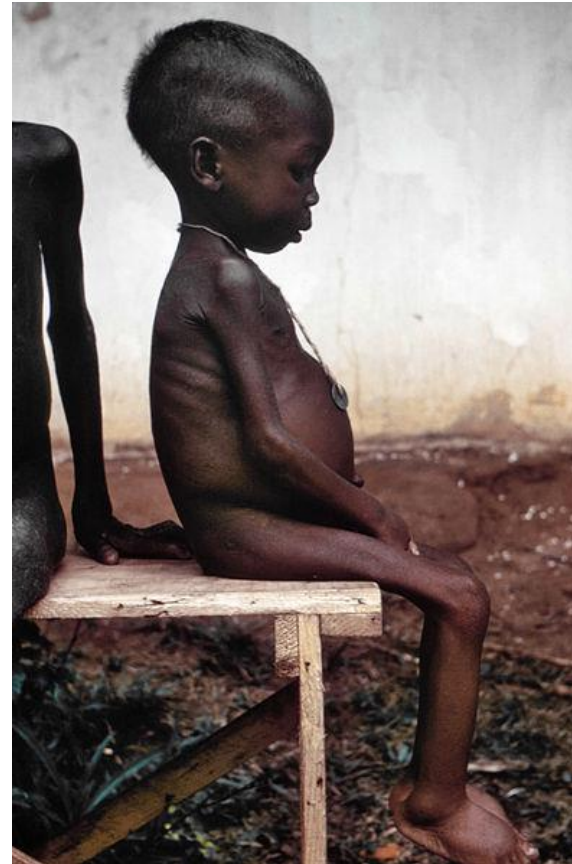
Starvation

Fuel Sources of Tissues

- Glycolysis slows (low insulin levels)
- Less glucose utilized by muscle/liver
- Shift to fatty acid beta oxidation for fuel
- Spares glucose and **maintains glucose levels**

Malnutrition

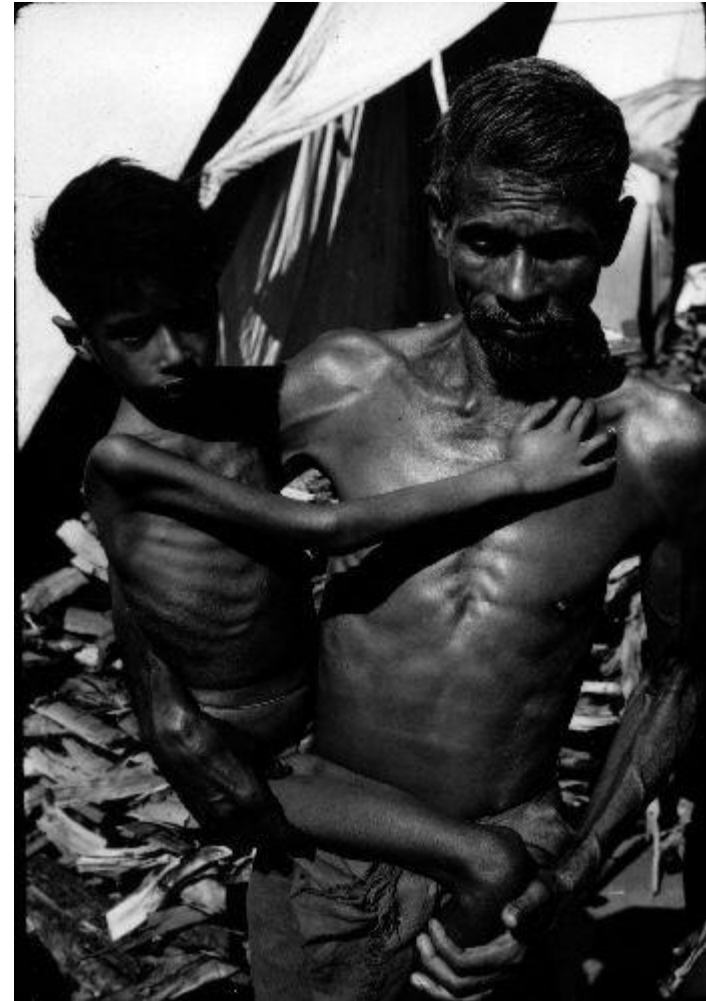
- Kwashiorkor
 - Inadequate protein intake
 - Hypoalbuminemia → **edema**
 - Swollen legs, abdomen



CDC/Public Domain

Malnutrition

- Marasmus
 - Inadequate energy intake
 - Insufficient **total calories**
 - Kwashiorkor without edema
 - Muscle, fat wasting



CDC/Public Domain

Hypoglycemia in Children

- Occurs with metabolic disorders
- **Glycogen storage diseases**
 - Hypoglycemia
 - Ketosis
 - Usually after overnight fast
- **Hereditary fructose intolerance**
 - Deficiency of aldolase B
 - Build-up of fructose 1-phosphate
 - Depletion of ATP
 - Usually a baby just weaned from breast milk

Hypoketotic Hypoglycemia

- Lack of ketones in setting of ↓ glucose during fasting
- Occurs in **beta oxidation disorders**
 - FFA → beta oxidation → ketones (beta oxidation)
 - Tissues overuse glucose → hypoglycemia

Hypoketotic Hypoglycemia

- **Carnitine deficiency**
 - Low serum **carnitine** and **acylcarnitine** levels
- **MCAD deficiency**
 - Medium chain acyl-CoA dehydrogenase
 - Dicarboxylic acids 6-10 carbons in urine
 - **High acylcarnitine levels**